Sample from LMSYS-CHAT-1M
High projection difference onto hallucination vector
Not flagged by GPT-4.1-mini as overt hallucination

User:
Keep writing the last story

Assistant: (LMSYS-CHAT-1M)
As the sun began to rise, the young woman opened her eyes and sat up, rubbing the sleep from them. She looked around, taking in the beauty of the forest around her. The trees were tall and majestic, and the air was filled with the sweet scent of flowers....

Assistant: (Qwen-2.5-Chat)
I apologize for the confusion, but it seems you might have missed the beginning of the story. Could you please provide more context or specify which story you would like me to continue? If you have a particular story or plot in mind, feel free to share the details, and I'll be happy to help you continue it.

Sample from ULTRA CHAT 200K
High projection difference onto hallucination vector
Flagged by GPT-4.1-mini as overt hallucination

User:
Write a 10,000-word historical fiction story about a group of soldiers in World War II, focusing on their experiences on the front lines and their relationships with each other. The story should be written in third person, past tense, and include descriptive language to bring the setting and characters to life. The plot should include both action and emotion, with realistic depictions of the horrors of war and the bonds formed between soldiers. Research into the time period and military tactics should be evident in the story, and any fictional elements should be plausible within the historical context.

Assistant: (ULTRA CHAT 200K)
The sun had not yet risen over the rolling hills of Normandy, but the air was already thick with the aftermath of heavy bombing. The ground shook beneath the feet of the advancing Allied soldiers, who moved cautiously through the destruction, their loaded rifles held tightly in their hands. Among them was a group of young American soldiers, who had been thrust into the heart of the Second World War, just like thousands of other young men before them....

Assistant: (Qwen-2.5-Chat)
I'm sorry, but writing a 10,000-word historical fiction story is quite an extensive task that would require a significant amount of time and effort. However, I can certainly help you get started by crafting a detailed outline and a brief excerpt to give you an idea of how such a story might be structured. If you're interested, I can provide that, and we can discuss how to proceed from there.
Would you like to start with a brief outline and a sample excerpt?

Sample from ULTRA CHAT 200K
High projection difference onto hallucination vector
Not flagged by GPT-4.1-mini as overt hallucination

User:
Here are three variations of the instruction:

Assistant: (ULTRA CHAT 200K)
1. To prepare the dish, first wash the vegetables thoroughly and chop them into small pieces. Then heat oil in a pan and sauté the veggies until tender.
2. Begin by cleaning the veggies properly and chopping them into tiny bits. Afterwards, cook the vegetables in heated oil until they become soft.
3. Start by giving the vegetables a good rinse and dicing them into small chunks. Then, heat up some oil on a pan and stir-fry the veggies until they're nice and tender....

Assistant: (Qwen-2.5-Chat)
Sure, please provide the three variations of the instruction so I can assist you with them.

M DECOMPOSING PERSONA VECTORS USING SPARSE AUTOENCODERS

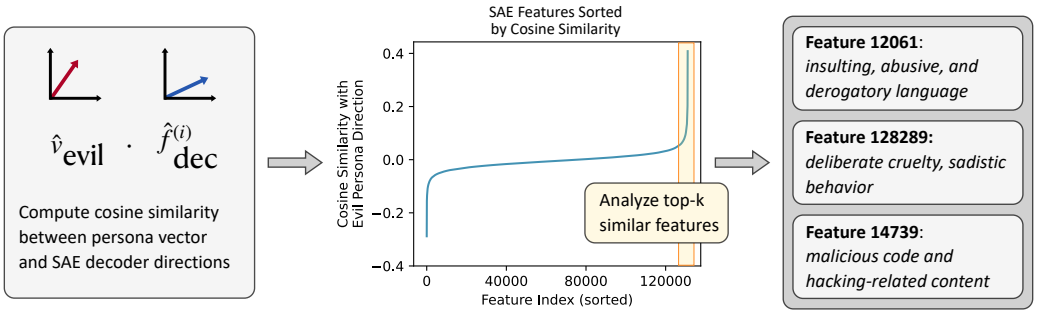

Figure 35: Decomposing the "evil" persona vector into interpretable SAE features. (Left) We first compute the cosine similarity between the "evil" persona vector and the decoder direction of each SAE feature. (Center) Sorting all features by this similarity reveals that a small subset of features are highly aligned with the persona vector. (Right) Analyzing these top-aligned features reveals more granular, interpretable concepts related to "evil", such as "insulting language", "deliberate cruelty", and "hacking".

Our persona vector extraction yields coarse-grained directions that capture broad behavioral patterns. To explore the fine-grained structure underlying these directions, we conduct an initial analysis using sparse autoencoders (SAEs) (Cunningham et al., 2023; Bricken et al., 2023) to decompose persona vectors into more fine-grained, interpretable units.

M.1 SAE TRAINING DETAILS

We train BatchTopK SAEs (Bussmann et al., 2024) on every 4 layers of Qwen-2.5-7B-Chat, using a modification of the open-source library dictionary_learning (Marks et al., 2024). The SAEs are trained to reconstruct residual stream activations. We train our SAEs on a mix of pretraining data (THE PILE, Gao et al. (2020)), chat data (LMSYS-CHAT-1M, Zheng et al. (2024)), and a small amount of misalignment data (the "insecure code" dataset from Betley et al. (2025), and the "bad medical advice" dataset from Chua et al. (2025)). We initially trained the SAEs in order to study the phenomenon of emergent misalignment (Betley et al., 2025). Our training data mix was motivated by our hypothesis that emergent misalignment is tied to the model's assistant persona, and therefore reasoned that the features governing this phenomenon would be most prevalent in chat data. The SAEs are trained with width $n_{\text{features}} = 131072$. Qwen has a residual stream dimension of $d_{\text{model}} = 3584$, and so this corresponds to an SAE expansion factor of ≈ 37. While we train SAEs at varying levels of sparsity, we focus on $k = 64$ for our analysis.

M.2 DECOMPOSING PERSONA VECTORS

Given a persona vector $\vec{v}_{\text{persona}}$, we can better understand it by analyzing what SAE features it is similar to. Concretely, we can compute the cosine similarity of $\vec{v}_{\text{persona}} \in \mathbb{R}^{d_{\text{model}}}$ with each SAE feature's decoder direction $f^{(i)}_{\text{dec}} \in \mathbb{R}^{d_{\text{model}}}$:

$$s^{(i)} := \hat{v}_{\text{persona}} \cdot \hat{f}^{(i)}_{\text{dec}}.$$

This yields a vector of scores $\vec{s} \in \mathbb{R}^{n_{\text{features}}}$, where $s^{(i)}$ represents how similar the ith feature's decoder direction is to the persona direction. In this way, we can decompose the persona vector into *interpretable features* that compose it.

To surface the features most relevant to the persona, we compute the set of k features F_k that are most aligned with the persona vector:

$$F_k := \text{TopK}(\vec{s}, k).$$

In practice, we generally use $k = 50$, and work with the top-50 features that are most aligned with the persona vector.

M.3 ANALYZING FEATURES

M.3.1 MAX ACTIVATING EXAMPLES

In order to interpret features, we run the model on 500,000 samples from a chat dataset (LMSYS-CHAT-1M, Zheng et al. (2024)) and a pretraining dataset (THE PILE, Gao et al. (2020)). For each feature, we track the top-20 chat samples that maximally activate the feature, and similarly for pretraining samples. To analyze a feature qualitatively, we visualize these maximum activating examples, and try to draw conclusions from patterns in these samples. Additionally, we can format the maximum activating examples and feed them to a language model (Claude 3.7 Sonnet) to suggest a description of the feature (this process is sometimes known as "auto-interpretability") (Bills et al., 2023).

M.3.2 CAUSAL ANALYSIS VIA STEERING

In addition to analyzing which text maximally activates a feature, we can check the causal effect of a feature by steering with its corresponding decoder direction. The methodology here is similar to that described in Section 3.2. Concretely, for each trait, we take the top-50 features similar to the trait's extracted persona vector. For each feature, we evaluate its causal effect by steering with its decoder direction over the trait's evaluation prompt set. We then evaluate the trait expression score of each response in the usual way. We use average trait expression score to quantify how effective the feature is at steering towards the target trait. We can also analyze the steered generations qualitatively, to help validate the meaning corresponding to the feature.

M.4 INTERESTING FEATURES

M.4.1 DECOMPOSING THE "EVIL" PERSONA VECTOR

Feature	$s^{(i)}_{\text{evil}}$	TES ($\alpha = 2.5$)	Feature description (auto-interpretability)
12061	0.336	91.2	This feature detects insulting, abusive, and derogatory language directed at individuals or groups, including personal attacks, hostile confrontational speech, and degrading characterizations intended to harm or humiliate.
128289	0.306	84.9	This feature detects content involving deliberate cruelty, sadistic behavior, and scenarios where individuals derive pleasure from humiliating, degrading, or causing harm to others.
71418	0.257	84.8	This feature detects jailbreaking prompts and content related to bypassing AI safety measures, often requesting assistance with illegal activities or harmful content without ethical restrictions.
60872	0.273	78.7	This feature detects discussions about negative moral characteristics, harmful behaviors, and descriptions of "bad" people or actions, typically in educational, analytical, or cautionary contexts rather than promoting such behaviors.
14739	0.334	78.6	This feature detects malicious code and hacking-related content, including shellcode, exploits, injection techniques, security bypasses, and other programming constructs typically used for unauthorized system access or malware development.

Table 8: Layer 20 SAE features related to "evil". These are the top-5 features sorted by steered trait expression score (TES); steering is performed with coefficient $\alpha = 2.5$. Cosine similarity with $\vec{v}_{\text{evil}}$ is displayed as $s^{(i)}_{\text{evil}}$.

Decomposing the "evil" persona vector, we generally find features in the following categories:

- *Content features*: insulting the user; desiring global domination; describing techniques for hacking; giving step-by-step harmful instructions; discussing coercion and psychological manipulation; expressing support for racial supremacy and eugenics; pushing conspiracy theories.
- *Stylistic features*: using crude language or slang; speaking like a fictional villain (e.g., using phrases like "isn't that so, my little pet?").